9. CROSS-SECTION & RETAINING WALL DRAWING

In this section, mortared stone retaining and shoring retaining wall drawings as cross-sectional and abutment structures will be explained.

9.1. CROSS-SECTIONS AND DRAWING

When designing a highway, calculations, and drawings are made on many subjects. The three most basic design processes in highway engineering include plan (horizontal design), profile (vertical design), and cross-section. Since the preparation of cross-sections is the basis for earthwork calculations, this is an important part of the design process. In order to draw a cross-section, the plan and longitudinal cros section of route must be prepared. A **cross-section is a** cross-section taken from any point of the road axis along a given line perpendicular to the road axis (this line is the cross-sectional line drawn by the rigth of way width in the plan view). In cross-sections, the platform (roadway) leveling line and slope lines are called the red line (design line), and the natural ground line consisting of fractures (landline) is called the black line. The cross-sectional area is calculated with the "Cross method" after finding the refraction points and coordinates (Y/X) of the characteristic points on these lines.

On the other hand, a typical cross-section shows in detail in Figure 9.1, the agreed standard dimensions and features of a road to be built. Typical cross-section, platform, coating, strip, banquette, cutting ditch, if any, refuge dimensions; transverse slope, slopes of cutting and embankment slopes; rigth of way width; Superstructure layers, thicknesses, and material types are specified according to the location (Figure 9.1). The scale of the typical cross-section is selected as 1/100. The line of ground (black line) is shown schematically on it. The geometry and values of the elements to be used in the highway assignment are given in the example figure below.

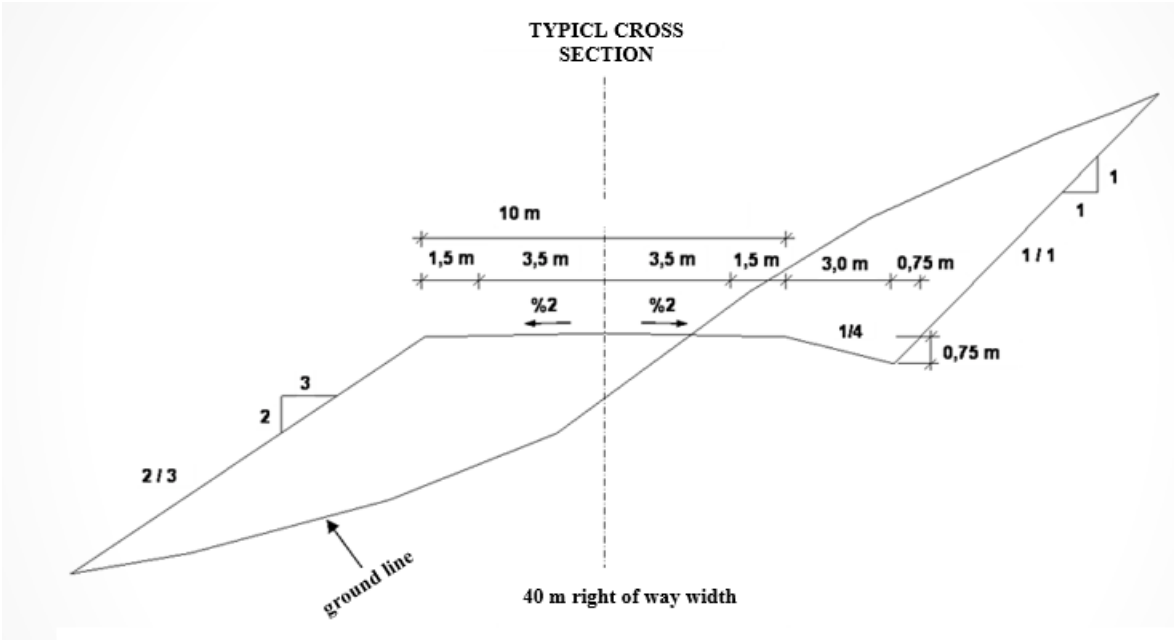


Figure 9.1 Typical cross-section and elements

Typical cross-section, plan, and longitudinal cross section of route are used when preparing a cross-section. To draw a cross-section, the following operations are followed in order:

1- At the beginning, since the type of earthwork (cutting, filling, mixed) of the cross-section to be drawn is not known exactly, the typical cross-section elements are drawn as a template (base) to show all cutting and filling situations on the right and left of the axis. The drawings to be made afterward are continued on this template (Figure 9.2). The horizontal and vertical scale of the template is the same and can be taken as 1/100 or 1/200, depending on the need. In the highway

assignment, 1/100 will be used first. If the letterhead does not fit on A4 paper, 1/200 can be used. If the drawing is done by hand, millimeter paper can be used conveniently.

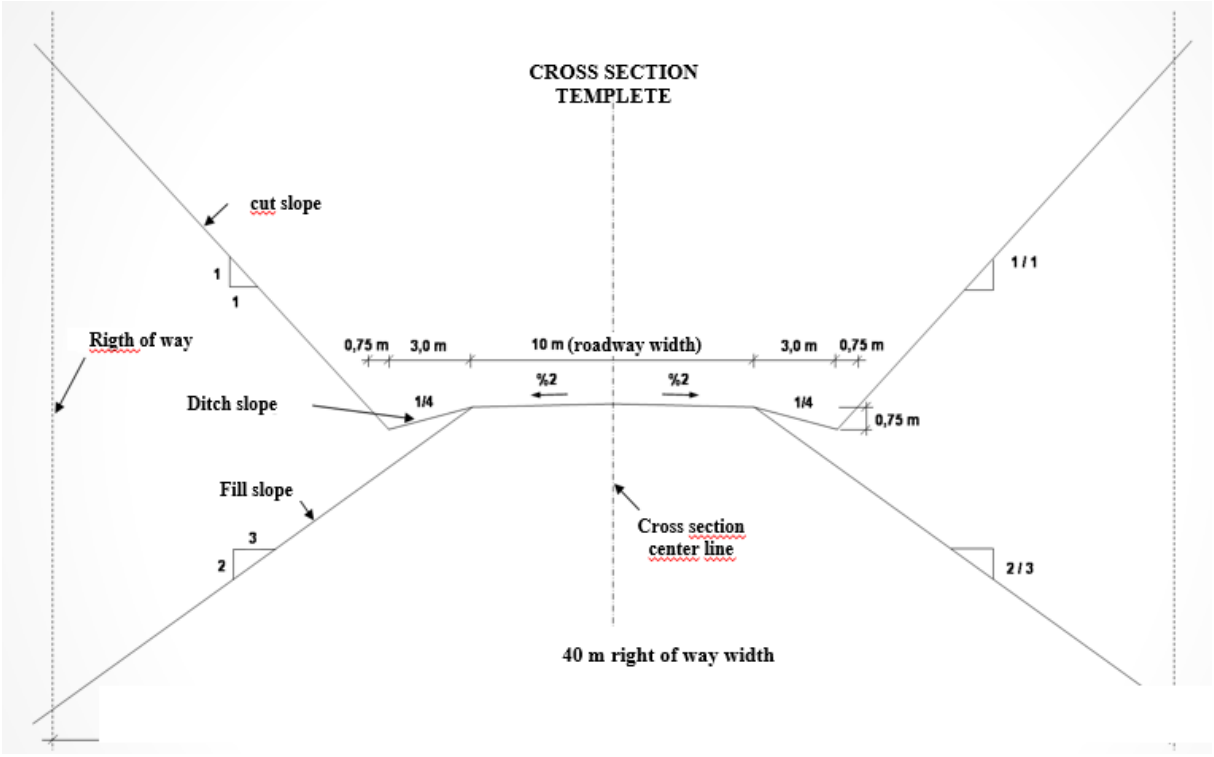


Figure 9.2 Cross-sectional drawing template

2- The point where the road platform intersects the axis is the red elevation in the longitudinal section. Therefore, the difference between black and red elevations in the profile is read. This length is passed through the scale and measured and marked from the point where the cross-sectional axis intersects the platform. The black elevation is above the platform in cut sections and on the bottom in fill sections. Thus, the black point of the cross-section on the axis (centerline) is determined. The coordinate of this point is also written next to it as (Y/X). Since it is on the axis line, it is shown as (Y=black elevation/0.00). The point where the axis intersects the platform is expressed as (Y = red elevation/0.00).

3- After the coordinates of the black and red points on the axis are determined, black elevation readings are made at critical points to define the landline on the right and left of the road axis according to the direction of roadway (In highway homework; black elevation readings should be done with a precision of 2 digits after the comma with a simple proportion calculation, not an eye decision). In the plan, critical points along the cross-sectional line drawn by the amount of rigth of way width; platform edges (on the right and left sides), the points where the line intersects the leveling curves, and the rigth of way boundary (on the right and left sides). Here, care should be taken to pass the distances read on the plan scale through the scale when moving to the cross-section. For example, in the Highway assignment, the scale of the plan is 1/2000, while the scale of the cross section is 1/100 (or 1/200). Lengths read at 1/2000 scale in the plan will be converted to 1/100 (or 1/200) when drawing the cross-section. The elevations and distances to the axis of all relevant critical points are marked by coordinating them to their respective places in the cross-section; Afterwards, it is combined in such a way that it follows each other and is broken. Thus, the condition of the land line in the cross section (black line) and therefore the type of cross section (full cut, full fill or mixed) that is the basis of the earthwork calculations is revealed (Figure 9.3-Figure 9.6).

4- It is checked whether the black line intersects the slope lines specified in the template (which are the other red lines of the cross-section). If the rigth of way cuts within its width and a closed area is formed with these cut points, the cross-sectional drawing has come to an end. The coordinates of these points are determined by a precise reading or calculation. Since the locations of the critical

points on the platform and trench will always remain constant as indicated in the typical cross-section, the red elevations and distances in these areas are easily calculated according to the red point on the axis. The Cross method calculates the area according to the coordinates of all the determined critical points. Cross-sections can be in full fill (Figure 9.3), full cut (Figure 9.4), cut-to-fill mixed (Figure 9.5), or fill-to-cut (Figure 9.6) types. However, in mixed types, the cutting and filling areas must be calculated separately. Since the part of the black line between the point where it cuts the slope surface and the expropriation boundary will no longer be used, it can be deleted at this stage, except for uses in some special cases. In addition, slope lines and trench drawings that are not needed should be deleted (see Figure 9.7- Figure 9.8c for an example).

5- On the upper part of the cross-sectional axis, in u order, the cross-sectional header information is written as follows:

Cross Section No:
Cross Section Km:
Red elevation:
Black elevation:

6- Cross-sectional areas found by Cross calculation are also written at the bottom according to their types (F y, Fd).

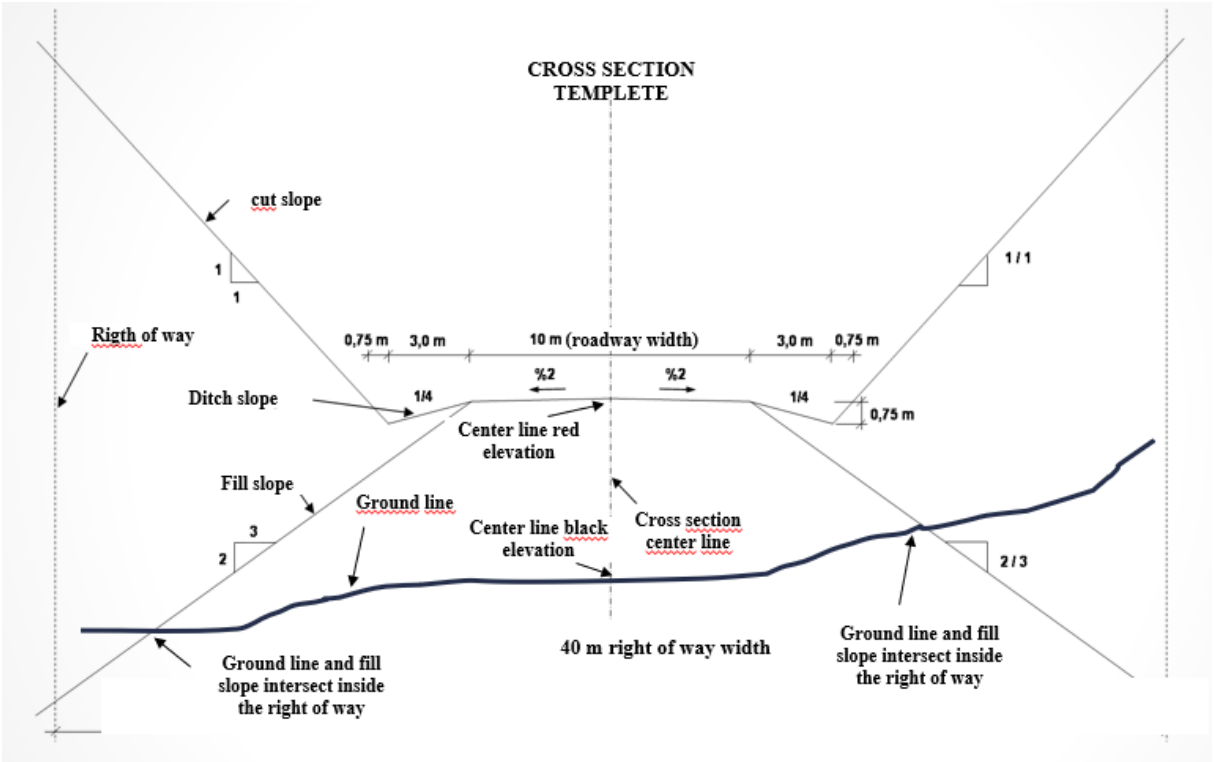


Figure 9.4 Full fill cross-section

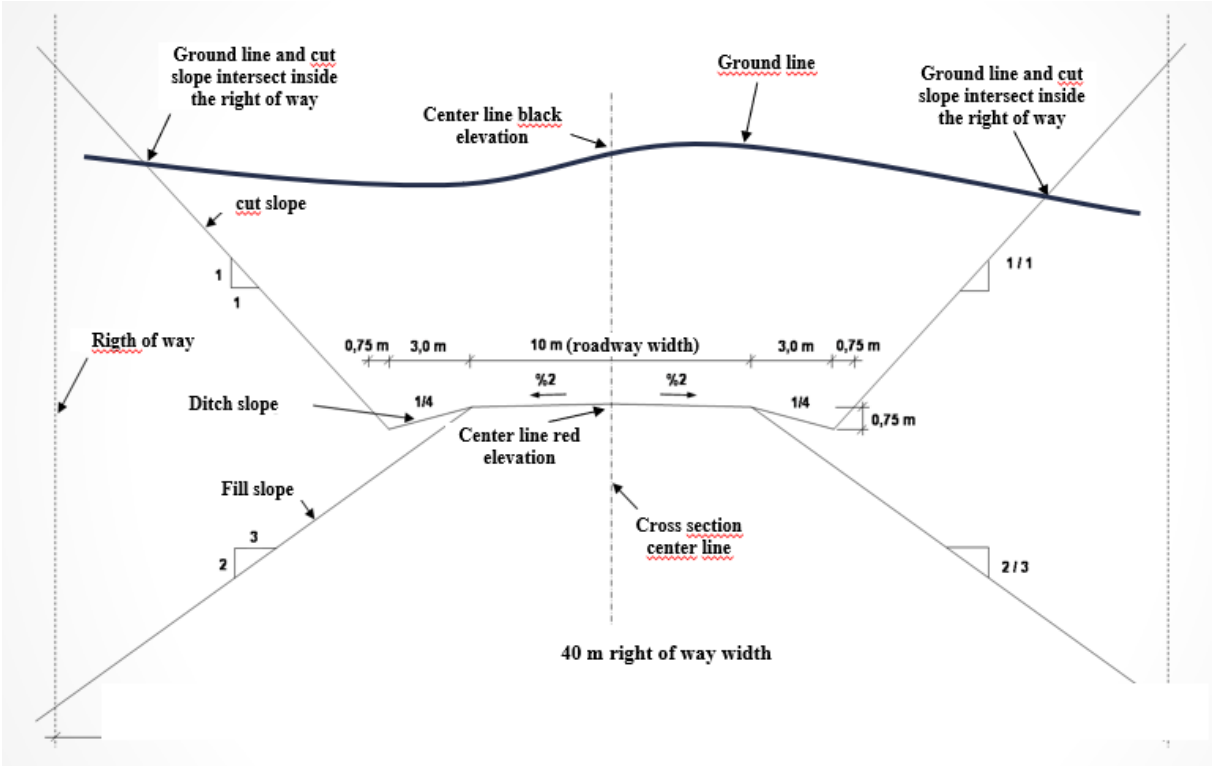


Figure 9.4 Full cut cross-section

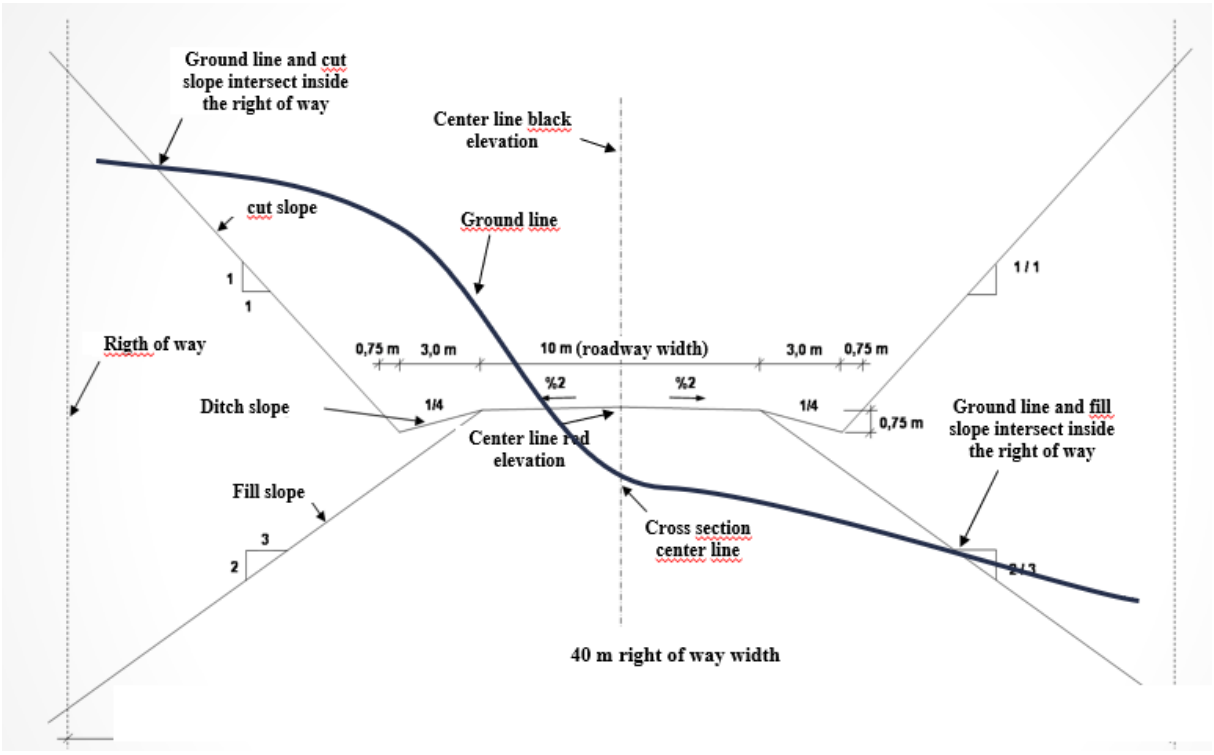


Figure 9.5 Mixed cross-section from cut to fill

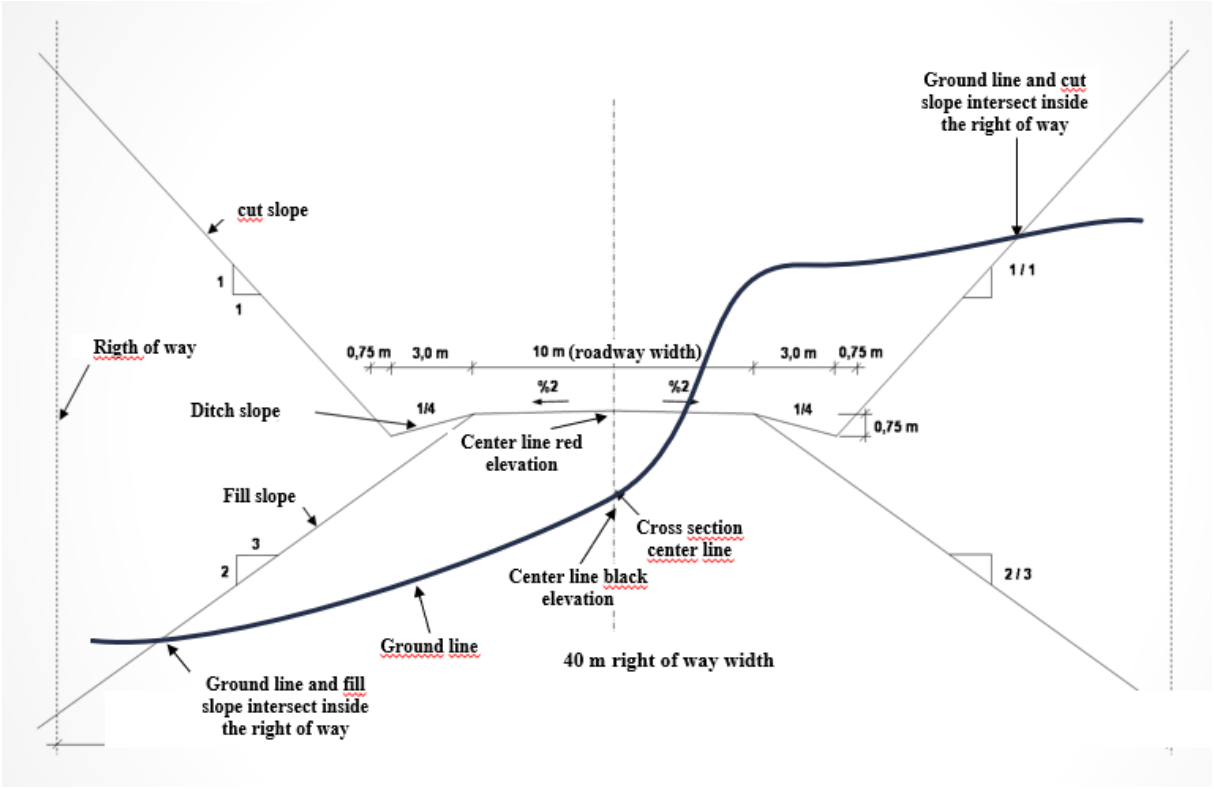


Figure 9.6 Mixed cross-section from fill to cut

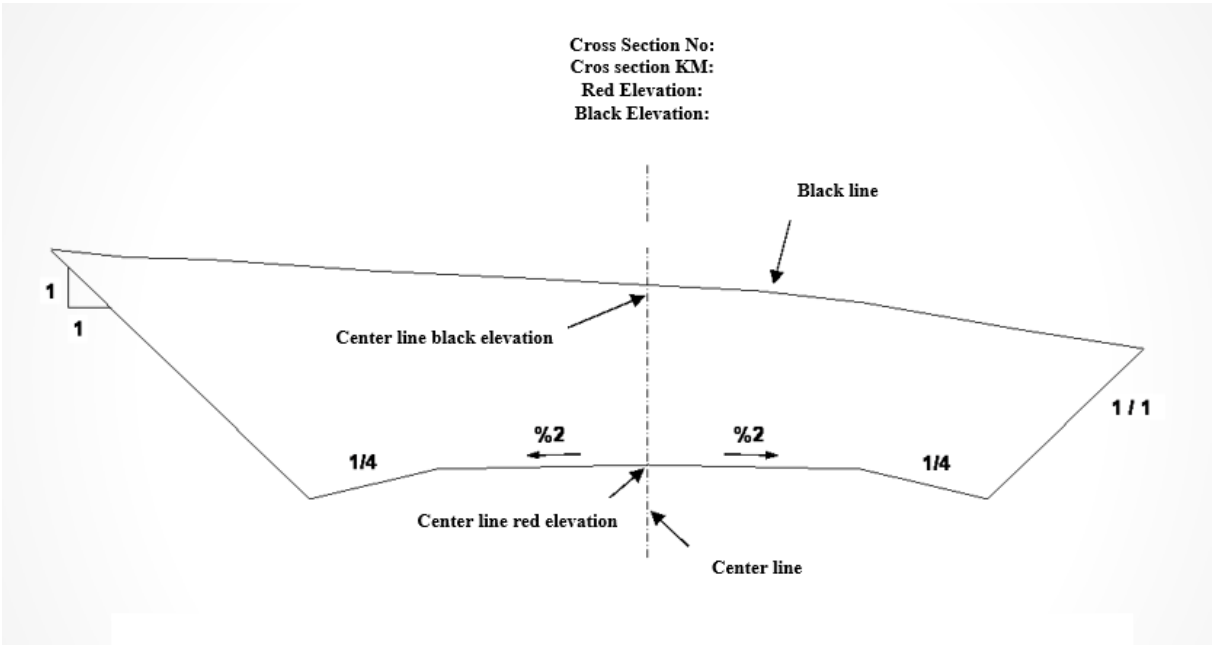


Figure 9.7 Full cut cross-section with unused stencil elements cleaned

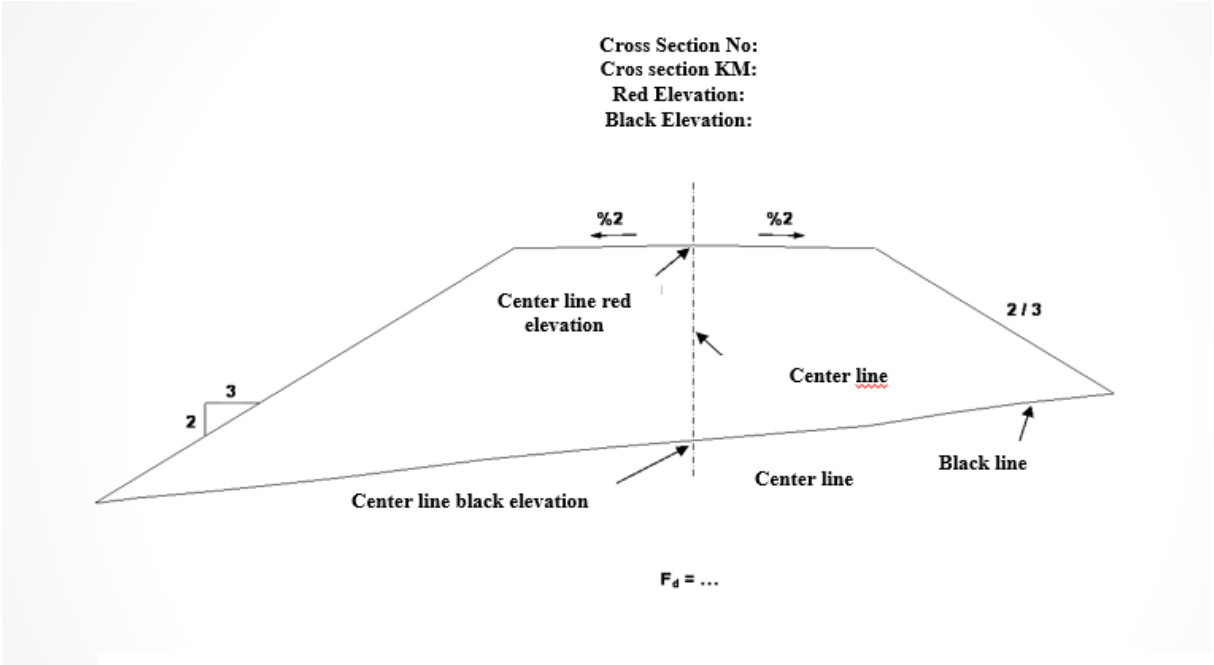


Figure 9.8a Unused stencil elements cleaned full fill cross-section

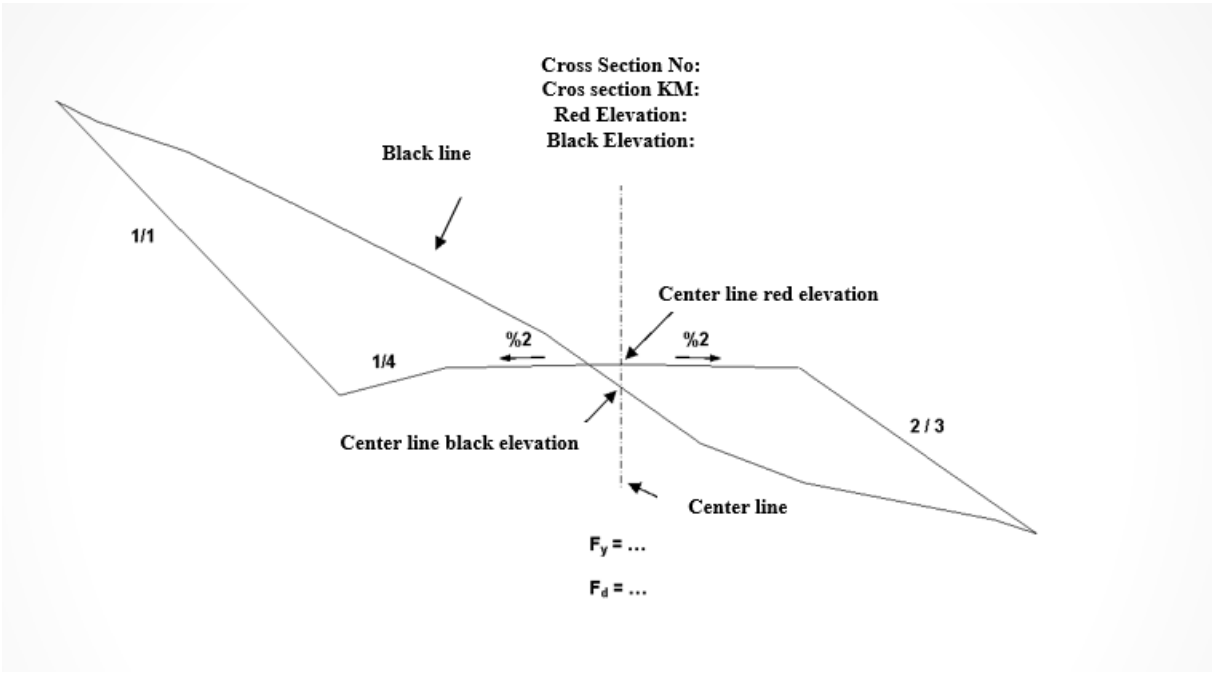


Figure 9.8b Unused stencil elements cleaned cut to fill mixed cross-section

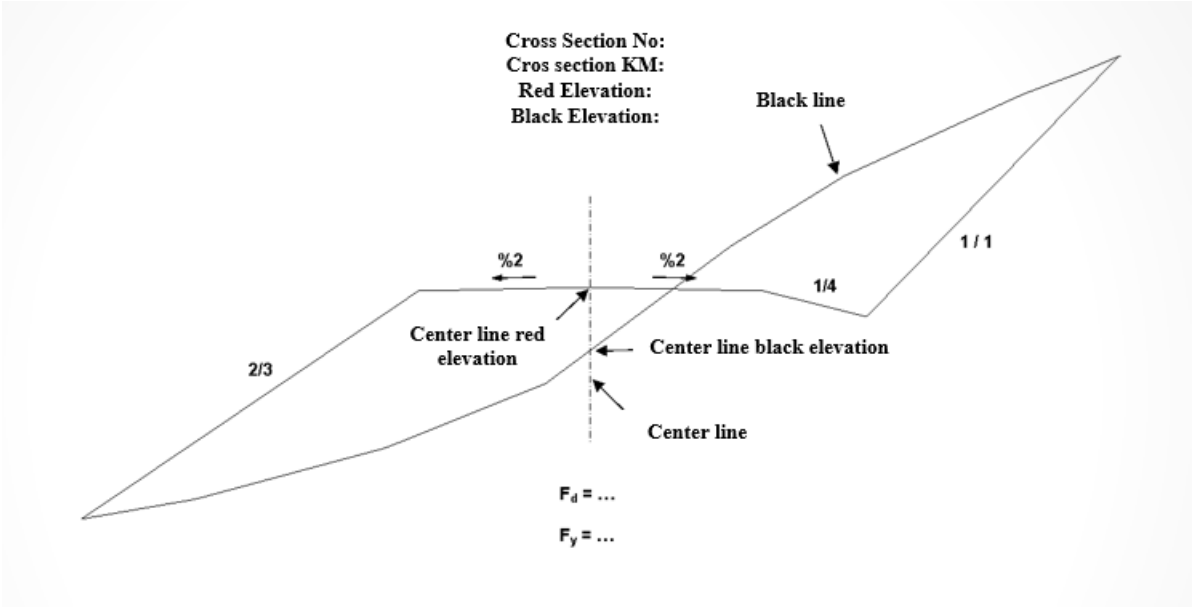


Figure 9.8c Unused stencil elements cleaned fill-to-cut mixed cross-section

7- If the black line does not intersect the slope lines specified in the template within the rigth of way width, no closed area exists (Figure 9.9- Figure 9.12). In this case, there is a possibility that sometimes there will be too much, sometimes almost endless earthwork. In such cases, there is a need to place a retaining wall in the filling part and a shoring wall in the cutting part. The following section will explain the topic of placing retaining/shoring walls in cross-sections.

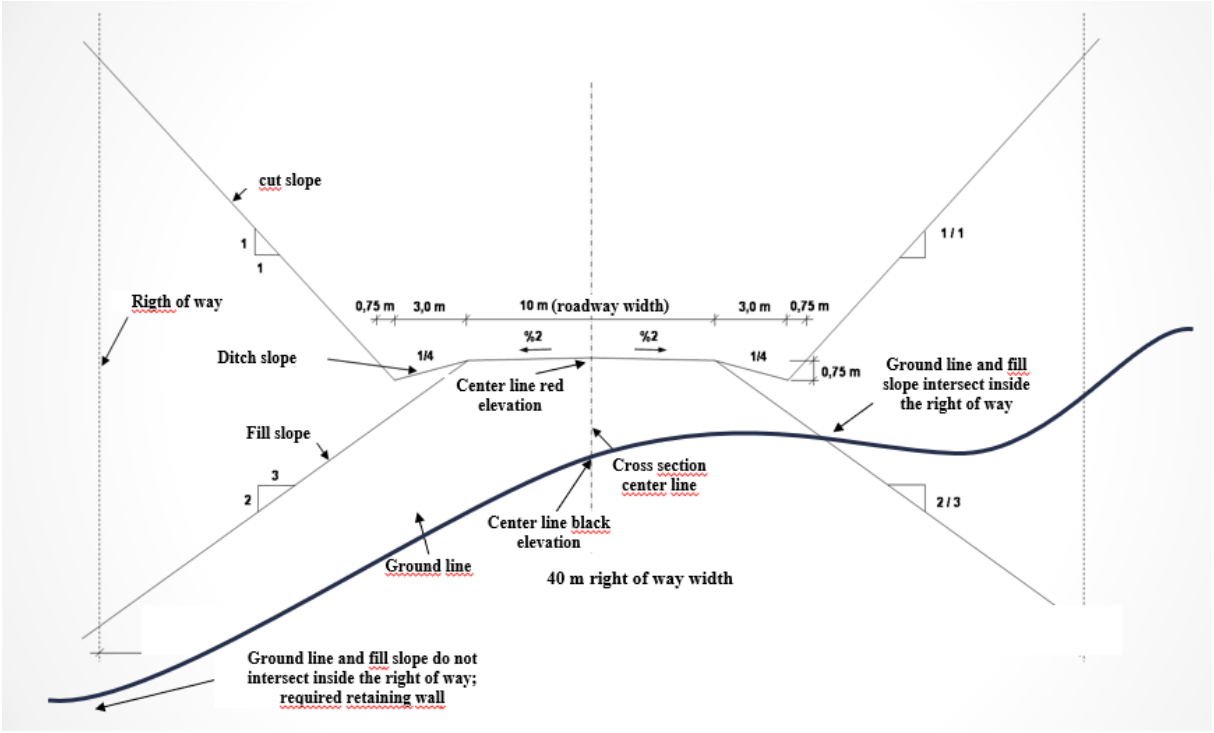


Figure 9.9 The need for retaining wall placement in full fill

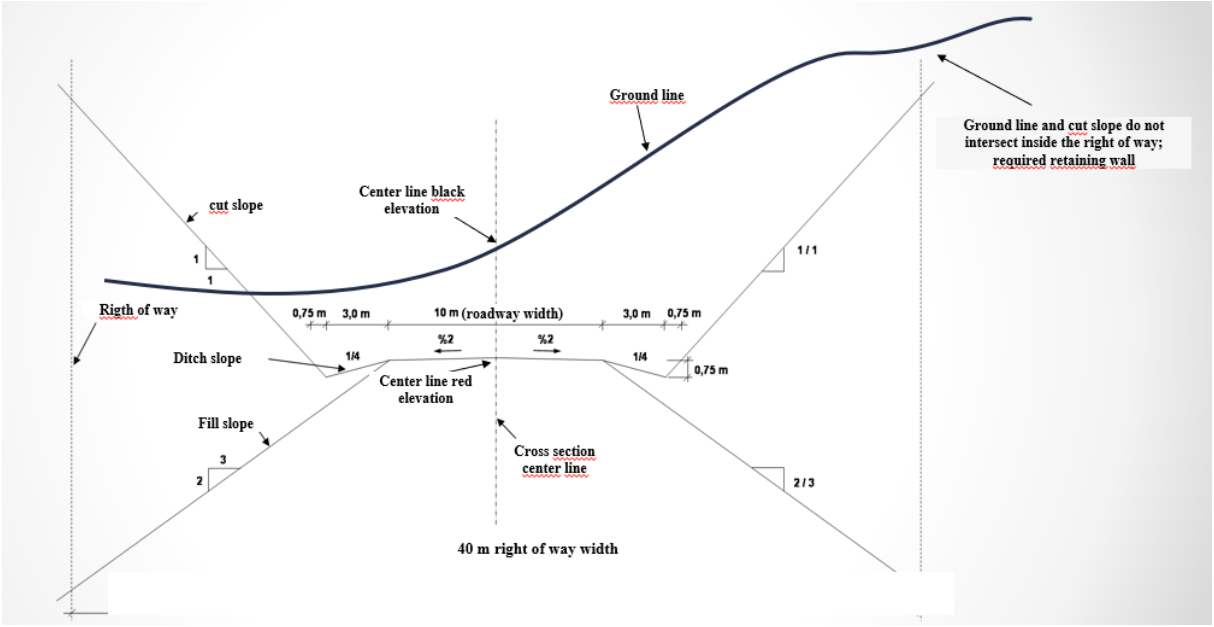


Figure 9.10 The need for shoring wall placement at full cut

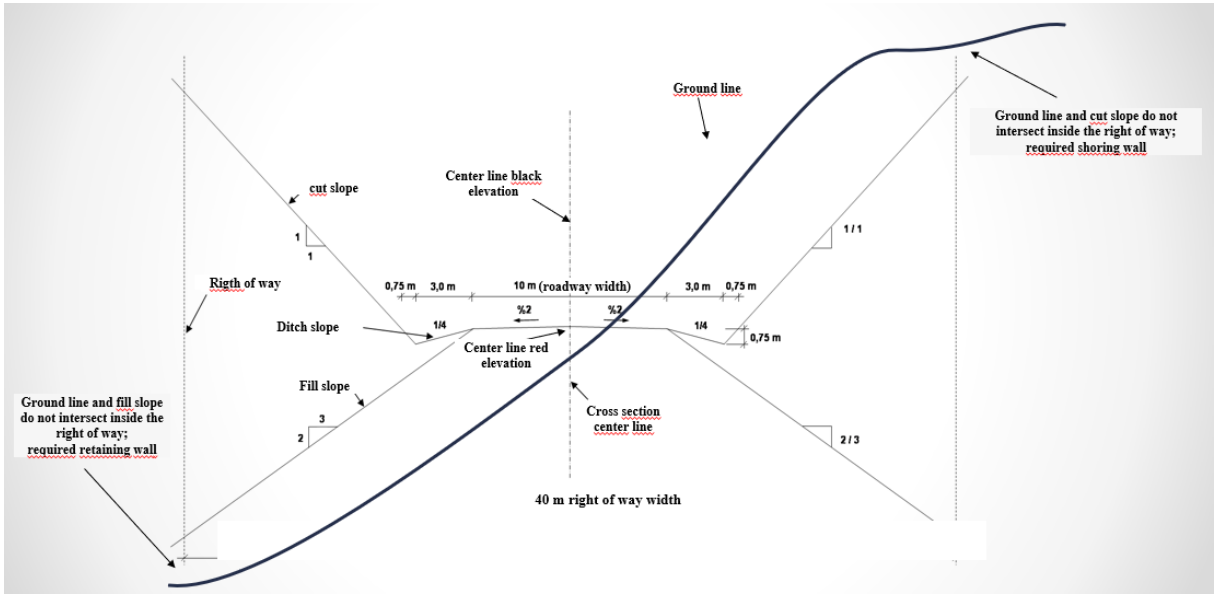


Figure 9.11 The need for retaining/shoring wall placement in mixed cross-section from fill to cut

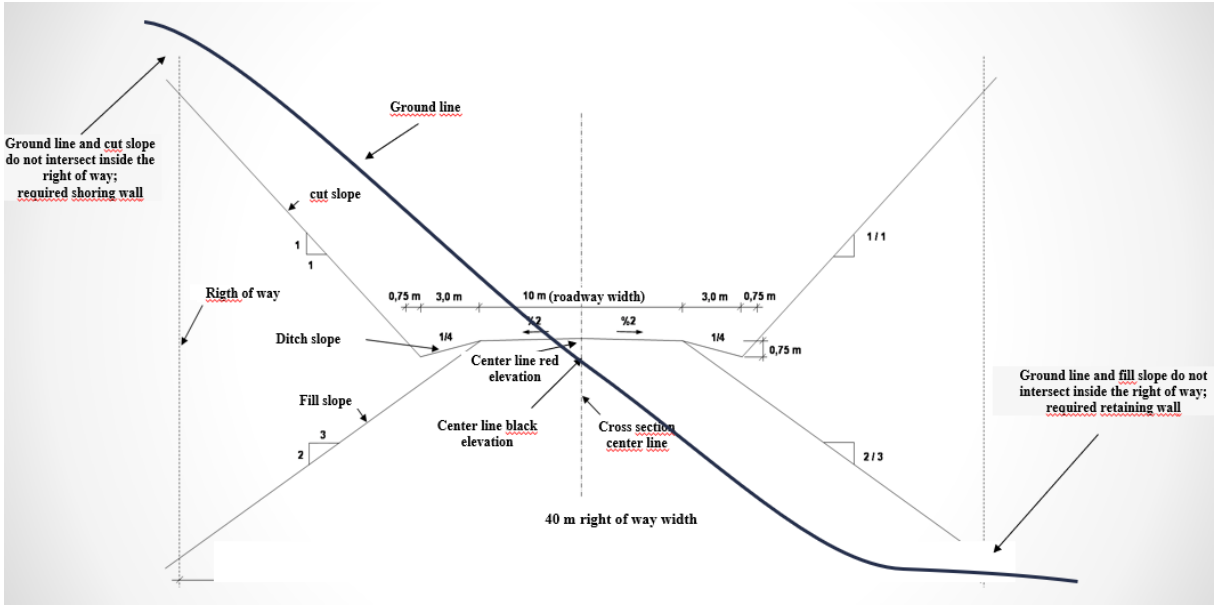


Figure 9.12 The need to place retaining/shoring walls in mixed cross section from cut to fill

9.2. RETAINING/SHORING WALLS (ABUTMENT STRUCTURES) AND DRAWING

Retaining/shoring walls (Supporting structures), in order to ensure slope stability in a highway design, to intersect embankment or cutting slopes with the ground, to avoid unwanted large filling and cutting production, to protect man-made structures on the roadside, as well as in sections where there are topographic necessities such as stream crossing, etc., on the wing walls of art structures such as bridges and culverts, in places where there is rigth of way problem, the cross-section that enables the road platform to be placed in accordance with the standards elements.The main type of retaining and shoring walls are gravity, cantilever, sheet piling and anchord.

Details about the placement of mortared stone retaining and shoring walls, which are widely used on highways, are given below.

9.2.1 Mortared Retaining/Shoring Wall

A mortared stone retaining/shoring wall is placed in the cross-section where the black line does not intersect the fill slope lines specified in the template within the rigth of way width or at a distance greater than 15 meters from the axis. The part of the retaining/shoring wall on the cross-sectional axis and the side of the fill area is defined as the inner surface, and the part where the filling slope is located in the opposite direction is defined as the outer surface. The sample drawings here will be made on two-lane undivided road sections, taking into account the soil ground conditions. To install a retaining/shoring wall, the following steps are followed in order:

Stage 1: A lateral distance between L = 7-10 m from the cross-sectional axis is selected. There is point N where the vertical projection of this intersects the fill slope (Figure 9.13). From this point, the upper surface of the retaining wall is determined by drawing a horizontal A-B line with a width of 0.50 meters, 0.30 meters out of the slope and 0.20 meters into the slope.

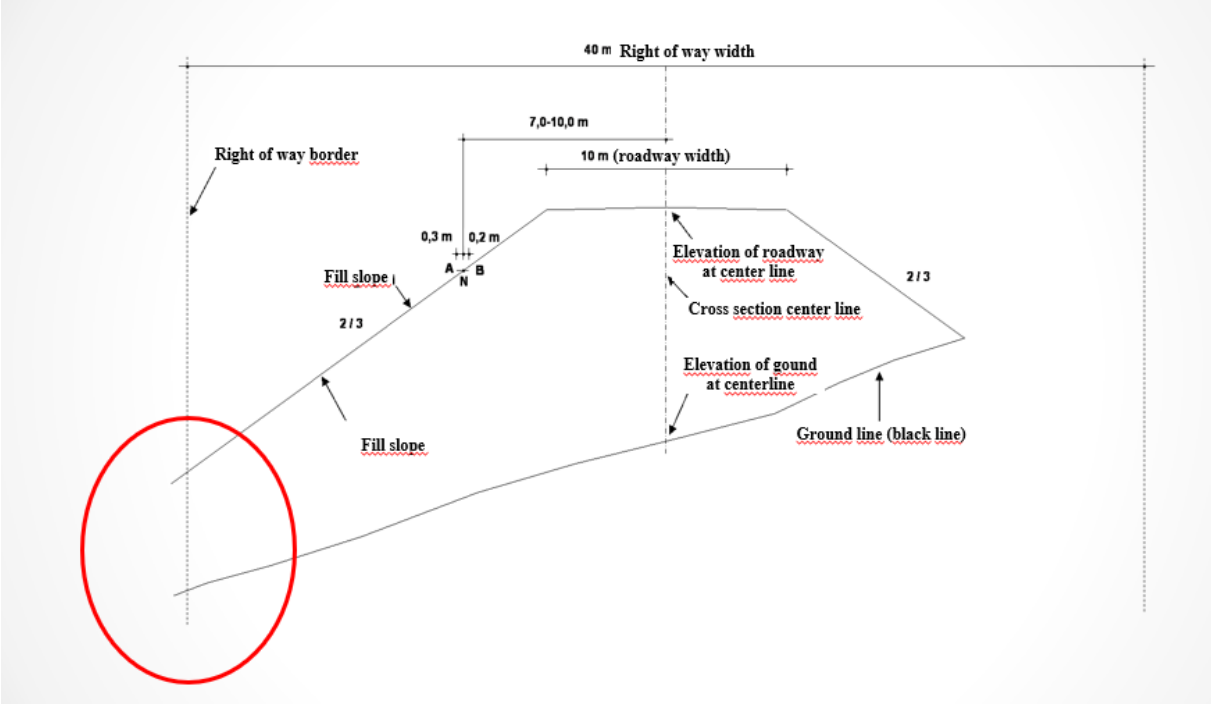


Figure 9.13 Retaining wall placement- 1. Stage: Determining the upper surface of the retaining wall

Stage 2: A line is drawn at an inclination of 5/1 (5 units vertical: 1 unit horizontal) from point A downwards and in the opposite direction of the cross-sectional axis (Figure 9.14). The following endpoint of this line is determined to be perpendicular to the ground (black line) at least 1 meter from the ground (point C). This line drawn is the outer surface of the retaining wall. The vertical distance of point C to the retaining wall top surface (h_T) is retaining wall height; The vertical distance (H) of the upper surface of the

retaining wall to the platform skirt is called the load height.

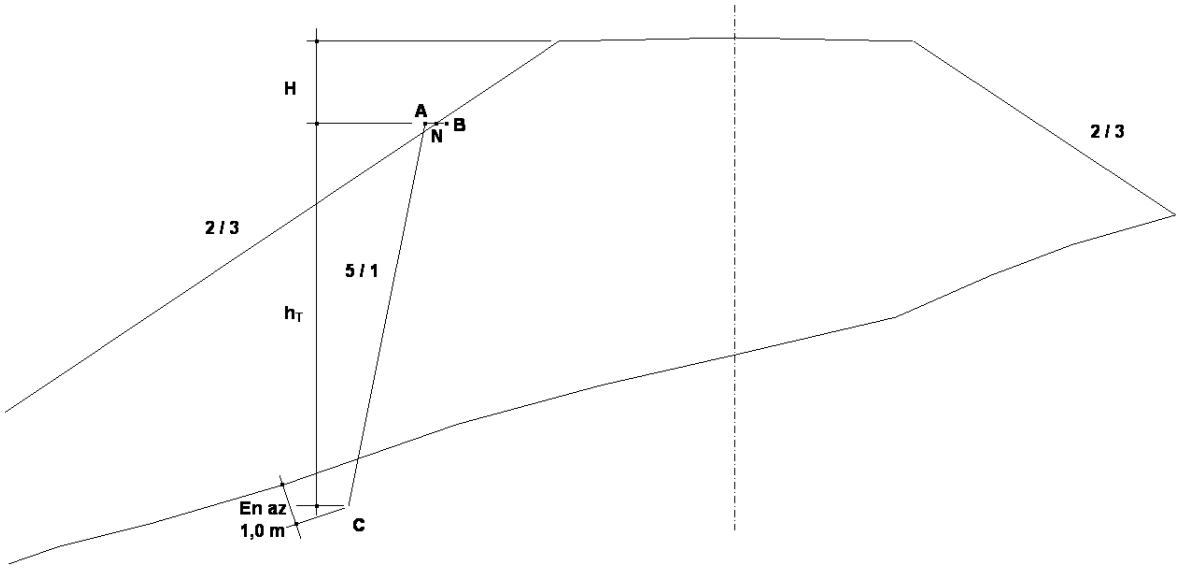


Figure 9.14 Retaining wall placement- 2. Stage: Determining the outer surface of the wall

Stage 3: From the lowest point of the retaining wall's outer surface line (point C), line C X is drawn in the direction of the cross-sectional axis and in the perpendicular direction (Figure 9.15). It is the bottom surface (base) of the retaining wall.

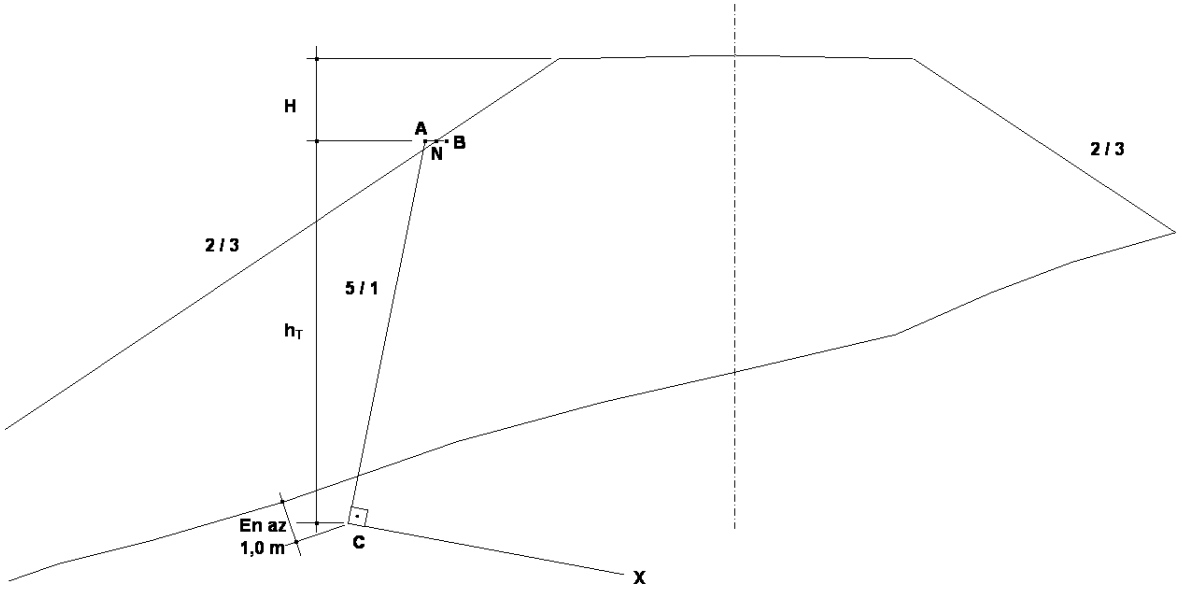


Figure 9.15 Retaining wall placement- 3. Stage: Determining the bottom surface of the retaining wall

Stage 4: A line B Z is drawn from the inner side of the upper surface of the retaining wall (point B) with a slope of 1/1 downwards and in the direction of the cross-sectional axis (Figure 9.16). This is the upper part of the inner surface of the retaining wall.

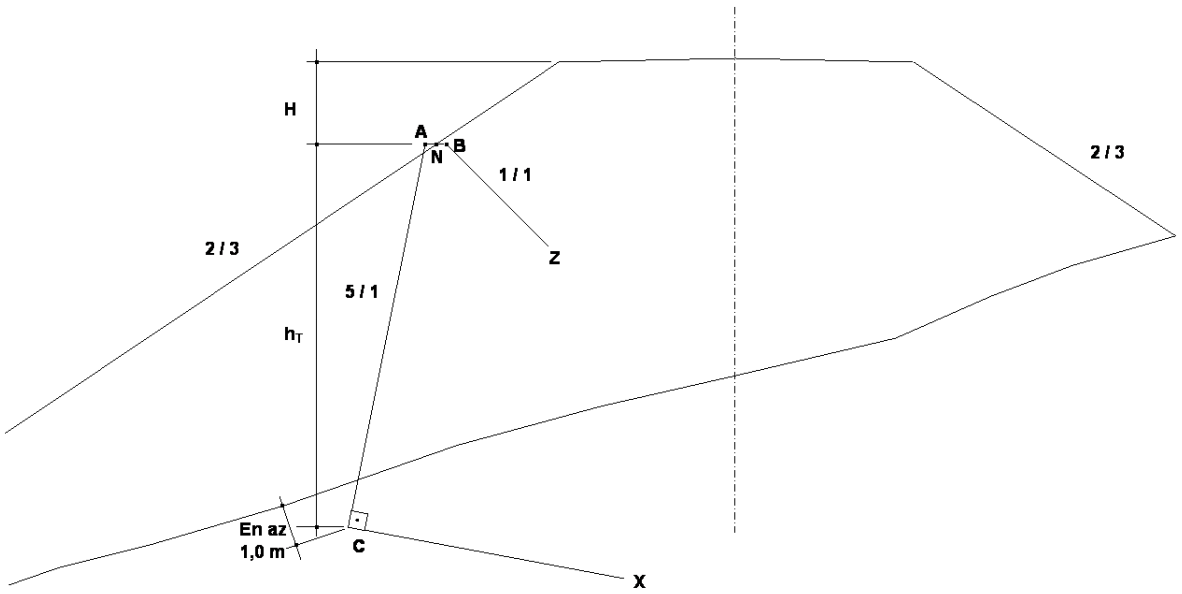


Figure 9.16 Retaining wall placement- 4. Stage: Determining the upper part of the inner surface of the retaining wall

Stage 5: The retaining wall height (h_T) and load height (H) values are determined in the table below, and the corresponding retaining wall thickness (K) is read here (Table 9.1). Interpolation is done for intermediate values. The determined retaining wall thickness is moved inward from the level of the perpendicular line descending from the outside of the upper surface of the retaining wall (point A) to the point where B intersects the line Z (point D) (Figure 9.17).

Table 9.1 Mortared stone retaining wall thicknesses

H, [m]	MORTAR RETAINING WALL THICKNESS (K), [cm]											
	hT (or h_K) Height, [m]											
	1	2	3	4	5	6	7	8	9	10	11	12
0	50	55	70	90	110	125	145	165	180	200	220	240
1	50	60	80	100	120	135	155	175	190	210	230	250
2	50	65	85	105	125	145	165	185	205	225	245	265
3	50	65	90	110	130	150	170	195	210	235	255	275
4	50	65	90	110	135	155	175	200	220	245	265	285
5	50	65	90	110	135	160	180	205	225	250	270	290
6	50	65	90	110	135	160	185	205	230	255	275	295
7	50	65	90	110	135	160	185	210	235	260	280	300
8	50	65	90	110	135	160	185	210	235	265	285	305
9	50	65	90	110	135	160	185	210	235	265	290	310
10	50	65	90	110	135	160	185	210	235	265	290	315
hT : retaining wall height to be taken in the soil												
hK : The height of the retaining wall to be taken on the cliff												
H : Load (slope on the retaining wall) height												

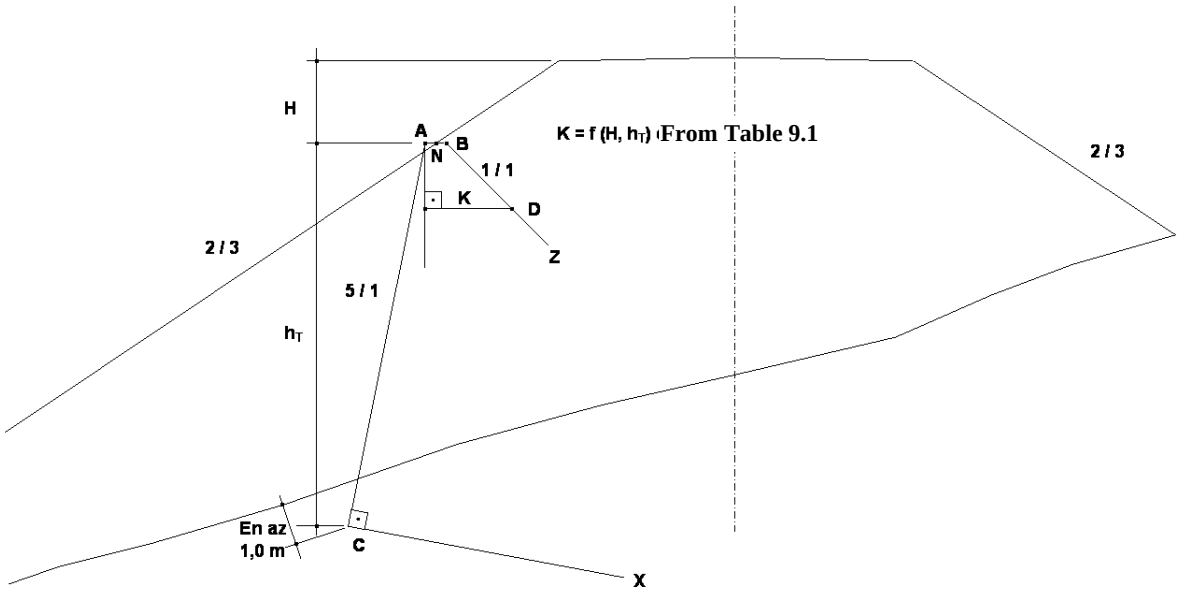


Figure 9.17 Retaining wall placement- 5. Stage: Finding the thickness of the retaining wall

Stage 6: A line is drawn vertically from point D downwards to point E. The endpoint of the DE line is determined as a distance of $0.6h_T$ vertically from the upper surface of the retaining wall (Figure 9.18). This line is the middle part of the inner surface of the retaining wall and is parallel to the perpendicular line descending from point A.

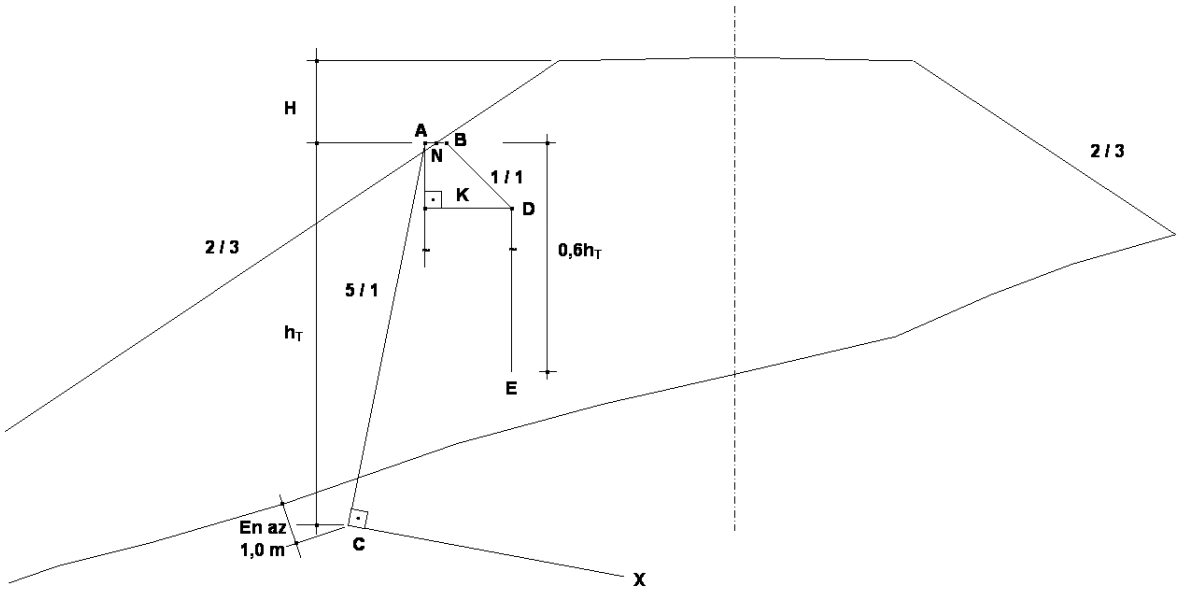


Figure 9.18 Retaining wall placement- 6. Stage: Determining the central part of the inner surface of the retaining wall

Stage 7: The parallel line EY drawn from point E downward to the AC line is joined by the line CX (base of the retaining wall) at point F (Figure 9.19). This is the lower part of the inner surface of the retaining wall. Thus, the process of placing the retaining wall is completed.

The point to be noted here is that if it is necessary to place a retaining wall in successive cross-

sections, the initial L distance must be selected as the same. In addition, all critical points of the retaining wall that show fracture are coordinated in order to be used in quantity surveying and cross-sectional area calculations (Figure 9.20). Cross-sectional areas with retaining walls are found without considering the retaining wall area. In order to drain the water that may be found from time to time in the embankment behind the retaining wall, drain pipes are laid at the cut points of the inner surface of the retaining wall with the landline. In addition, in the direction of the upper surface of the retaining wall, it is useful to lay these drain pipes at intervals of approximately 3 meters, although it is determined according to the need.

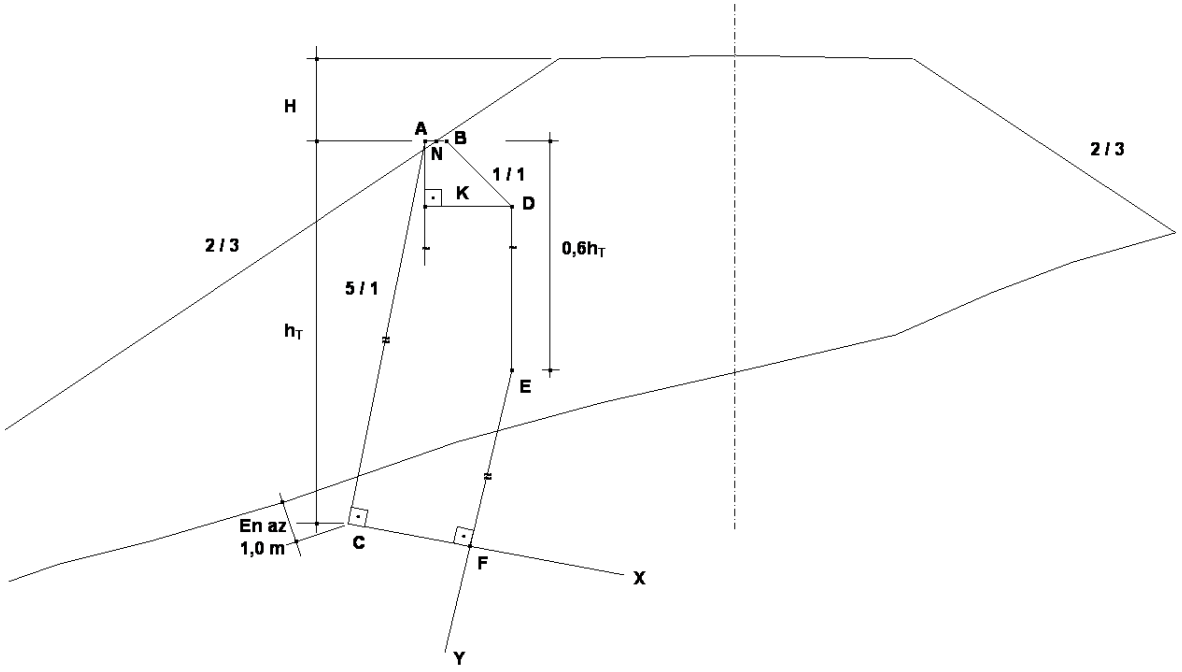


Figure 9.19 Retaining wall placement- 7. Stage: Completion of the inner face of the retaining wall

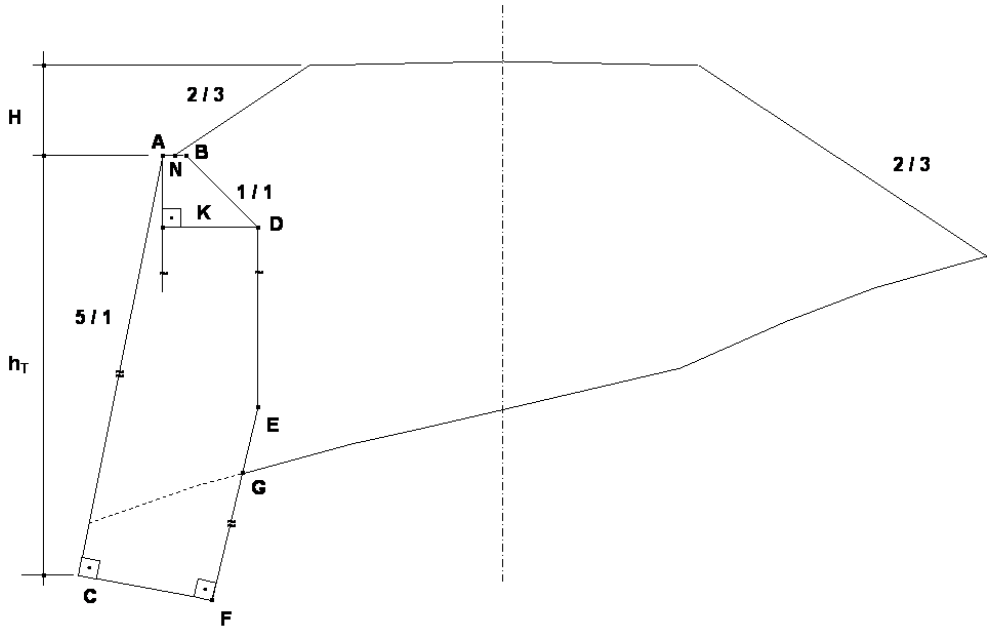


Figure 9.20 Coordinate of critical points of the retaining wall

9.2.2 Mortar Shoring Wall

The mortared stone shoring wall is placed in the cross-section where the black line does not intersect the cutting slope lines specified in the template within the expropriation width or at a distance greater than 15 meters from the axis. The part of the shoring wall on the side of the cross-sectional axis and the cutting area is defined as the outer surface, and the part where the cutting slope is located in the opposite direction is defined as the inner surface. The sample drawings here will be made on two-lane undivided road sections, taking into account the soil ground conditions. In principle, the shoring wall is designed like a retaining wall, except for one issue. The difference is in determining the location of the shoring wall at the first stage. Accordingly; The location of the shoring wall is not subject to a choice as in the retaining wall, but is the bottom point of the cutting ditch. The drawing of the line representing the outer surface of the shoring wall starts from the bottom of the cleavage ditch. In this case, the following steps are followed in order to place the shoring wall:

Stage 1: The landline is estimated by drawing a line with a slope of 5/1 (5 units vertical: 1 unit horizontal) in the opposite direction of the cross-sectional axis from the bottom point of the ditch (point A) where the cutting ditch slope and the cutting slope intersect (Figure 9.21). The following end point of this line is determined to enter the ground perpendicular to the bottom of the ditch by at least 1 meter (point B). This line drawn is the outer surface of the shoring wall.

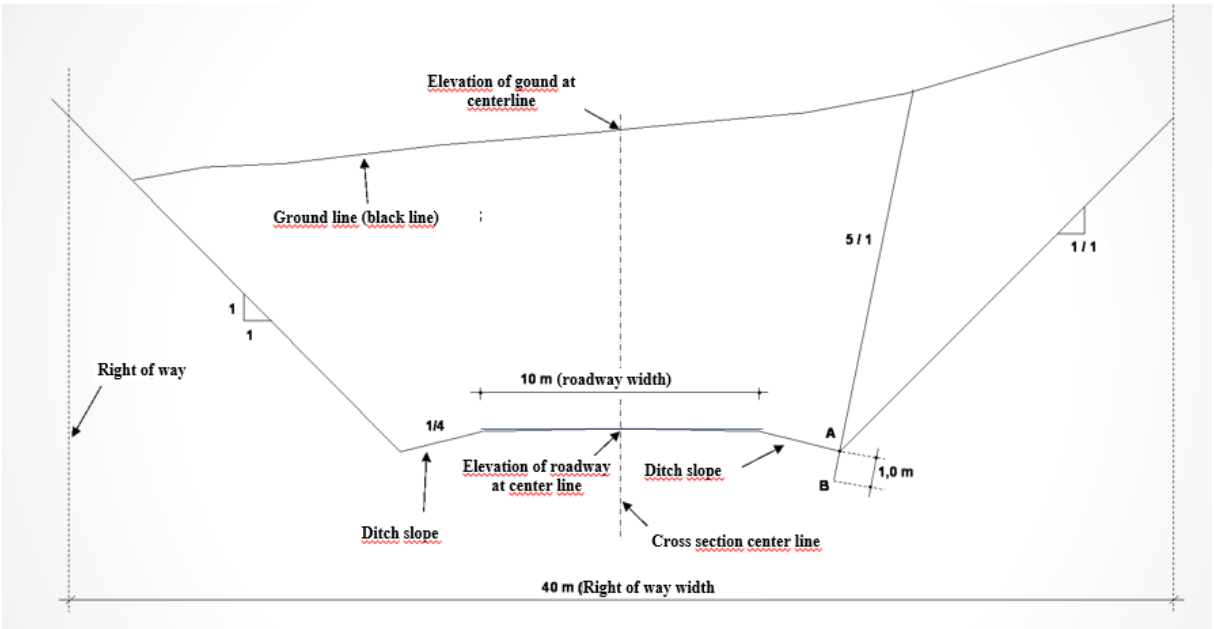


Figure 9.21 Shoring wall placement- 1. Stage: Determining the outer surface of the shoring wall

Stage 2: After the outer surface line of the shoring wall crosses and crosses the ground, the top surface of the shoring wall with a width of 0.5 m is determined by drawing a horizontal line segment in the opposite direction of the cross-sectional axis, 0.30 meters outside the landline and 0.20 meters inside the landline (Figure 9.22). That is, the outside of the shoring wall upper surface is point C; point N, where the upper surface of the shoring wall intersects the line of ground; The horizontal line segment of the CD is drawn, with the inside of the upper surface of the shoring wall being the D point.

Stage 3: From the lowest point of the shoring wall outer surface line (point B), in the opposite direction of the cross-sectional axis and in the perpendicular direction, the line B_X is drawn (Figure 9.23). That's right, it's the bottom surface (base) of the shoring wall.

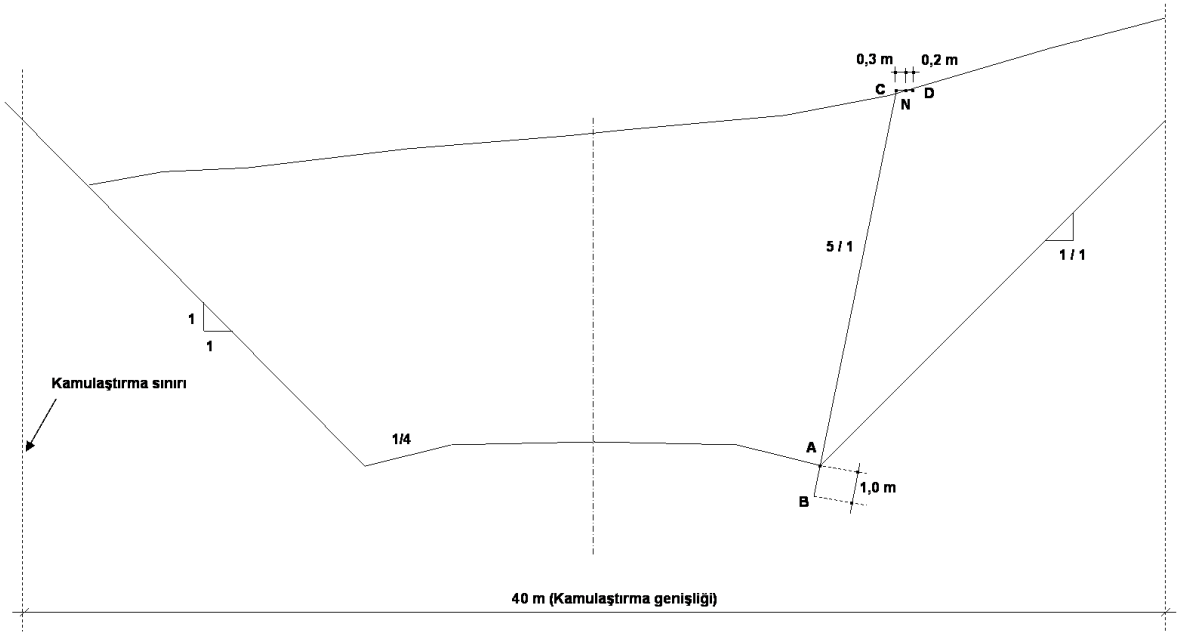


Figure 9.22 Shoring wall placement- 2. Stage: Determining the upper surface of the shoring wall

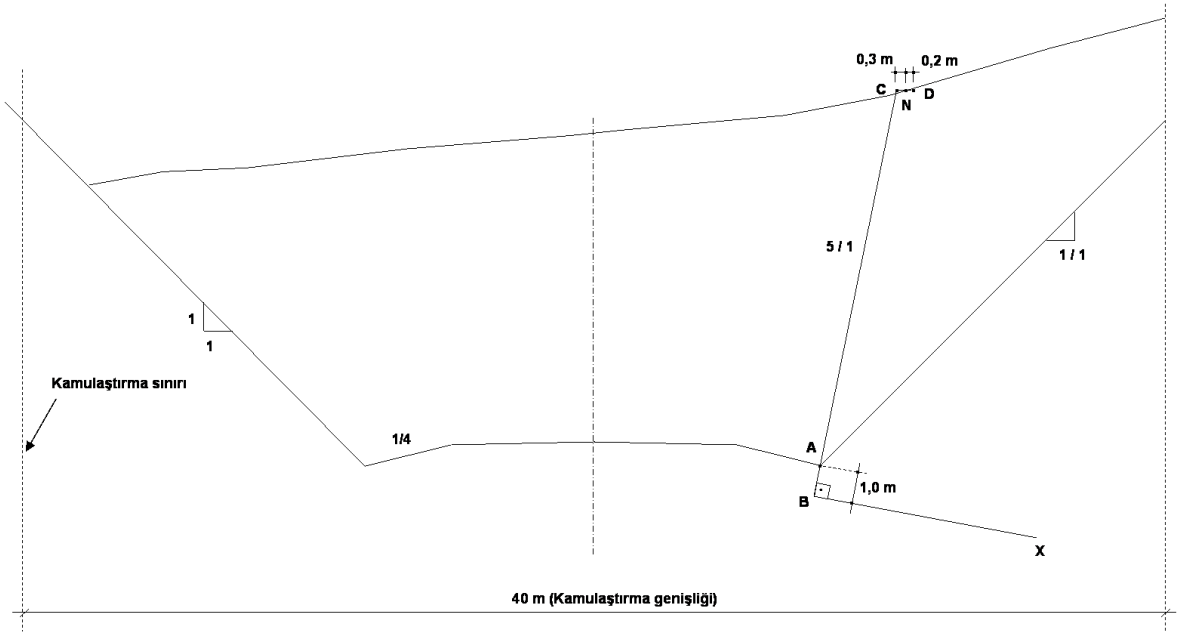


Figure 9.23 Shoring wall placement- 3. Stage: Determining the bottom surface of the shoring wall

Stage 4: The shoring wall load height (HT) and shoring wall height (h_T), which are the subject of the thickness chart (Table 9.2), are determined. To do this, the cutting slope line and the landline are first joined at point M, except for the expropriation width, to determine the shoring wall load height (Figure 9.24). The vertical distance of this point measured from the shoring wall top surface line CD line is the load height. The vertical distance between the bottom of the cutting trench (point A) and the line of the shoring wall surface CD is measured as the shoring wall height.

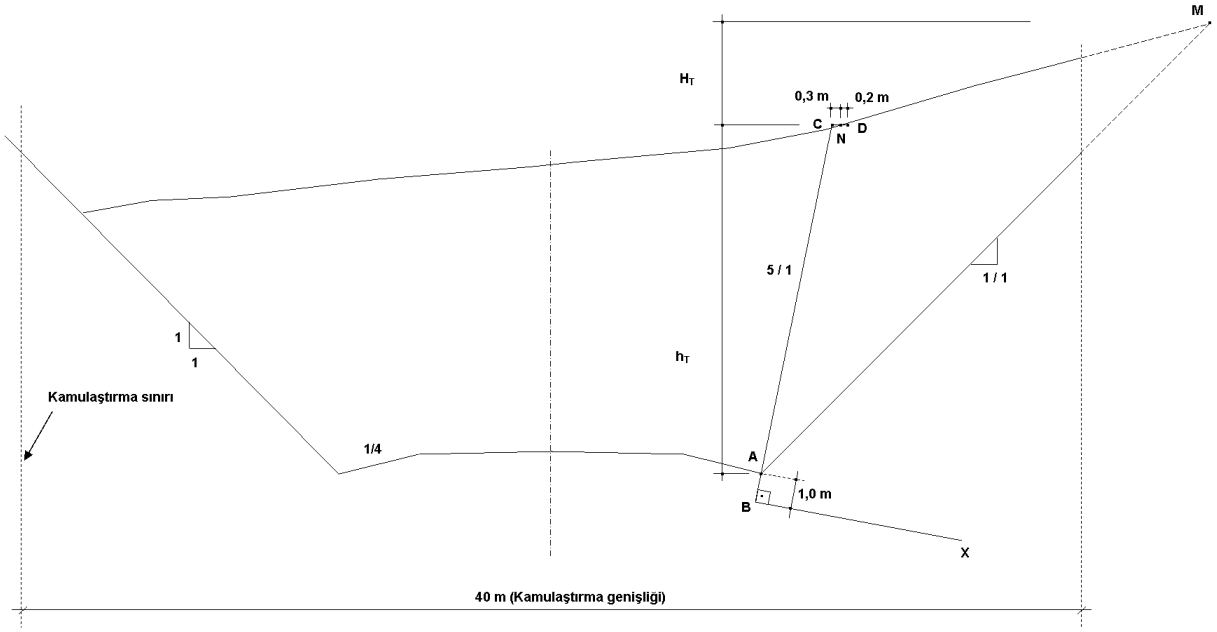


Figure 9.24 Shoring wall placement- 4. Stage: Determining the height of the shoring wall

Stage 5: A line DZ is drawn from the inner side of the upper surface of the shoring wall (point D) with a slope of 1/1 downwards and in the opposite direction of the cross-sectional axis (Figure 9.25). This is the upper part of the inner surface of the right shoring wall.

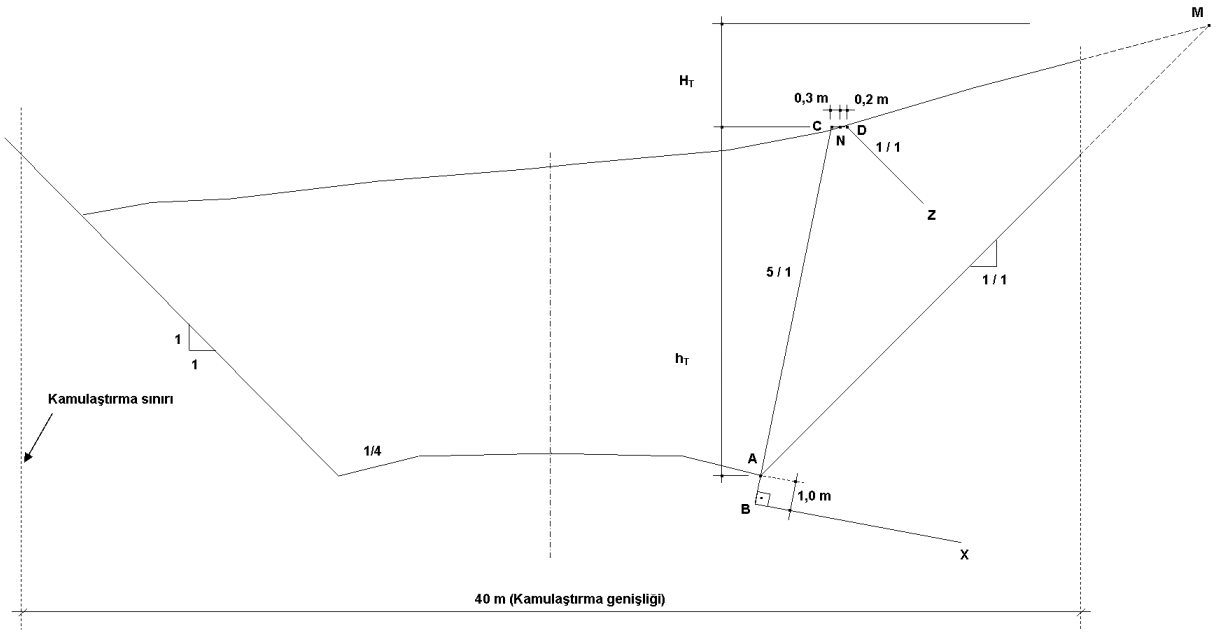


Figure 9.25 Shoring wall placement- 5. Stage: Determining the upper part of the inner surface of the shoring wall

Stage 6: The shoring wall height (h_T) and load height (H_T) values are determined in the table below, and the corresponding shoring wall thickness (K) is read here (Table 9.2). Interpolation is done for intermediate values. The determined shoring wall thickness is moved inward from the line of the

perpendicular line descending from the outside of the upper surface of the shoring wall (point C) to the point where D intersects the line Z (point E) (Figure 9.26).

Table 9.2 Thicknesses of mortared stone shoring walls

HT, HK, [m]	MORTARED SHORING WALL THICKNESS (K), [cm]											
	h_T (or h_K) Height, [m]											
	1	2	3	4	5	6	7	8	9	10	11	12
0-1	55	60	65	80	95	115	130	150	170	185	205	225
2	55	60	65	80	100	120	135	155	175	190	210	230
4	55	60	70	85	105	125	140	160	180	200	220	240
6	55	60	70	90	110	130	150	170	190	210	230	250
8	55	60	75	95	115	135	155	175	195	215	235	255
10	55	60	80	100	120	140	160	180	200	220	240	265
12	55	60	80	105	125	145	165	190	210	230	250	270
16	55	60	80	105	130	150	175	195	215	240	265	285
20	55	60	80	105	130	155	180	200	225	245	270	295
h_T : Wall height to be taken in the soil												
h_K : The height of the wall to be taken on the cliff												
HT : Load height in soil (slope on the wall)												
HK : Height of the load (slope on the wall) on the rock												

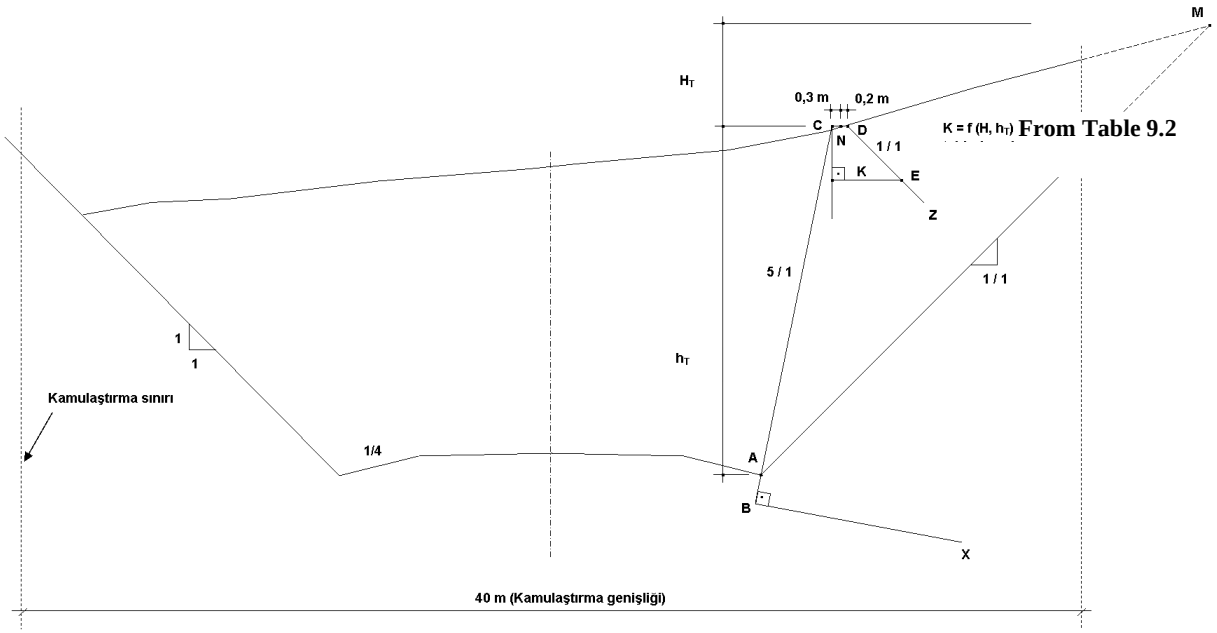


Figure 9.26 Shoring wall placement- 6. Stage: Finding the thickness of the shoring wall

Stage 7: A line is drawn vertically from point E to point F. The end point of the EF line is determined as a distance of 0.6hT vertically from the upper surface of the shoring wall (Figure 9.27). This line is the middle part of the inner surface of the shoring wall and is parallel to the perpendicular line descending from point C.

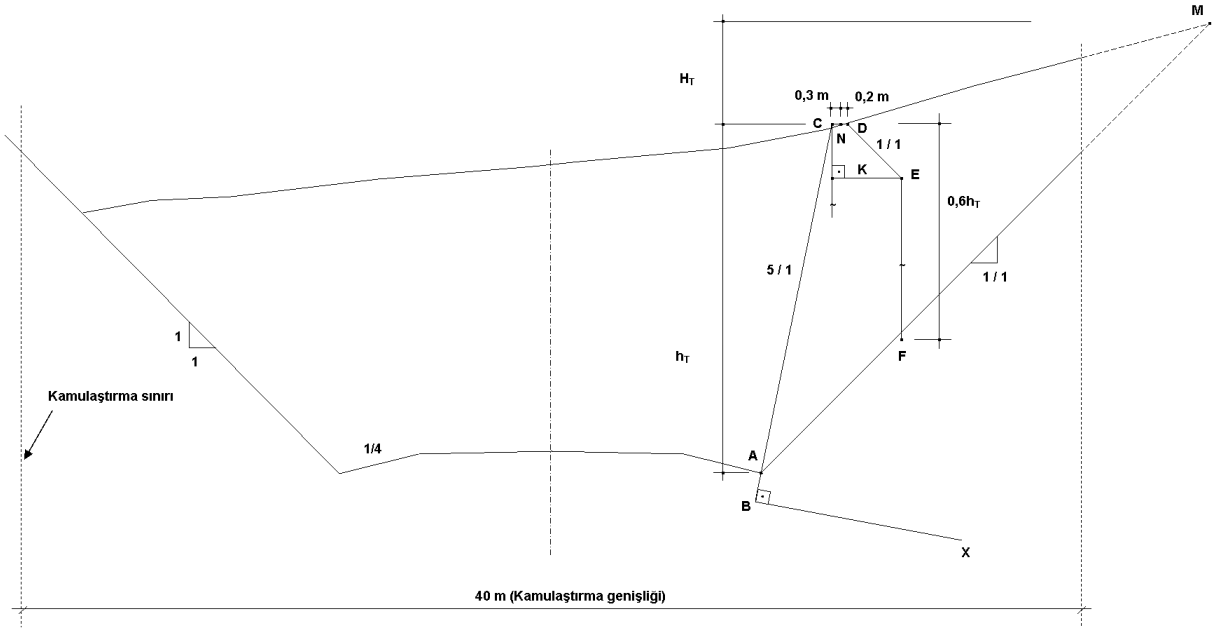


Figure 9.27 Shoring wall placement- 7. Stage: Determining the central part of the inner surface of the shoring wall

Stage 8: The parallel line F Y drawn from point F downward to the CB line is joined by line BX (the base of the shoring wall) at point G (Figure 9.28). This is true for the lower part of the inner surface of the shoring wall. Thus, the process of placing the shoring wall is completed.

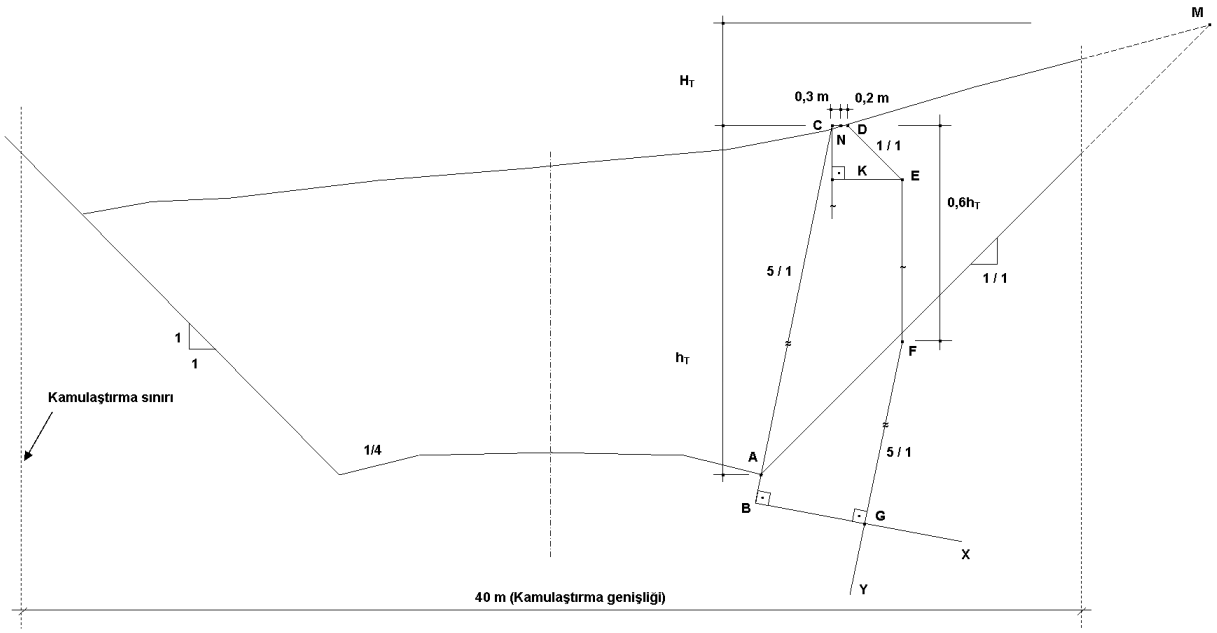


Figure 9.28 Shoring wall placement- 7. Stage: Completion of the inner face of the shoring wall

In order to be used in quantity surveying and cross-sectional area calculations, all critical points of the shoring wall that show fractures are coordinated (Figure 9.29). Cross-sectional areas with shoring walls are found by considering the shoring wall area.

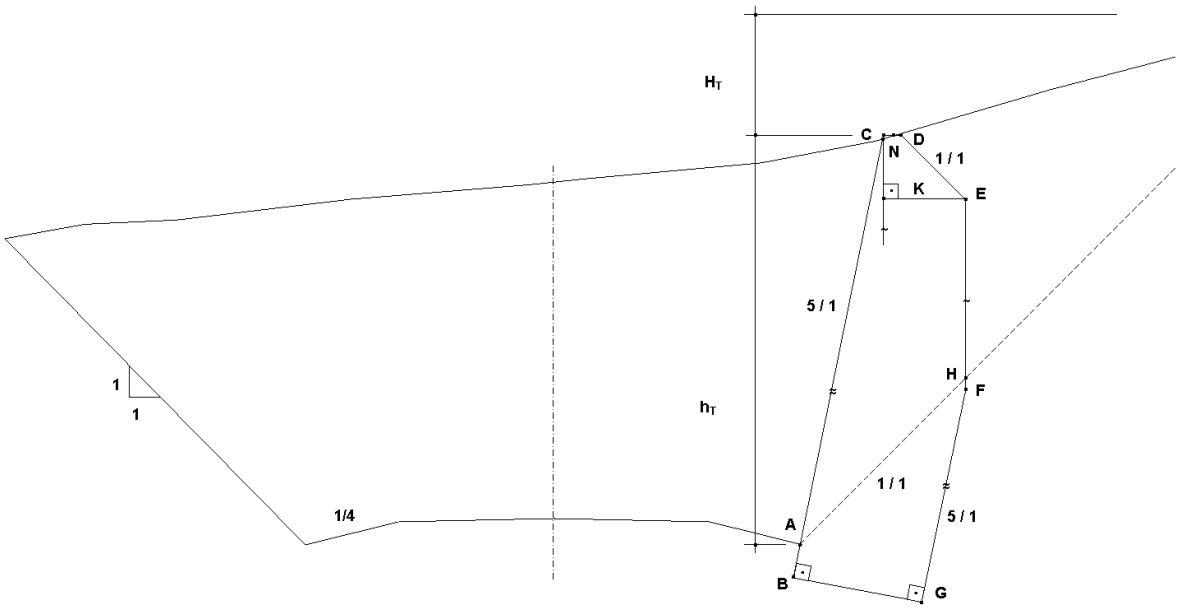


Figure 9.29 Coordinate of critical points of the shoring wall